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STEADY-STATE FREQUENCY ANALYSIS OF THE "LLC" RESONANT INVERTER FOR INDUCTION HEATING.

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Abstract: In this paper, the main steady-state frequency properties of the LLC circuit are described. The advantage of its use on induction heating applications is proved, in contrast with the SRI (series resonant inverter). This analysis will allow the future design of a frequency control strategy for the inverter. Finally, a simple design algorithm is proposed, which satisfies the power specifications and limits the stress current in the switches.

INTRODUCTION

The LLC equivalent circuit for induction heating is shown in *fig.1*, also called R.T. (Resonant Transformation) in [1]. In the schematic, the heating coil with load (metallic piece) is represented by Z_h (series asociation of L_p and R), while L_s and C are two reactive components appended to the previous to conform the R.T.

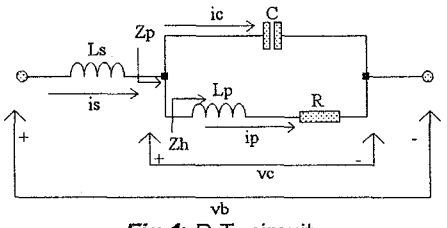

Fig.1: R.T. circuit.

The R.T. impedance is attacked by a rectangular voltage waveform, from -*V* to *V*, generated by the switching of transistors and diodes in a full/half bridge. This waveform can be expresed in the Fourier summatory form, which, using the superposition principle, allows to study each harmonic independently.

The aim of the paper is to obtain the several frequency functions which describe the behavior of the circuit, i.e., we want to detemine the answer of the circuit in front of an input type,

$$v_b(t) = v\cos(\omega t) \tag{1}$$

which could represent any of the harmonics of the input voltage.

When the applied voltage is square shaped, as happens in the frequency control of the resonant converters, the amplitud of the *n-th* harmonic, v_n, is:

$$v_n = \frac{4V}{n\pi} \tag{2}$$

being v_1 the fundamental harmonic amplitude. Due to that, to the circuit properties themshelves, and to the foreseen working conditions, the fundamental harmonic will cause the most intense response of the circuit and it will be the main cause of power dissipation.

THE PARALLEL CIRCUIT C//R-Lp

The parallel circuit $C//R\text{-}L_p$ is shown in *fig.2*. This circuit is the responsible of the ratio between the current amplitudes i_s and i_p, and it determines the frequency at which the maximum of this ratio is achieved. Since i_s is the current through the switches, and i_p is the cause of the power dissipation on the load, the desired situation is such that i_s is small but i_p is very intensive, in such a way that it´s posible to work with high power but low current stresses in the transistors and diodes. This R.T.´s faculty of amplify the current through the heating coil makes the R.T. structure preferable over the SRI for the induction heating applications.

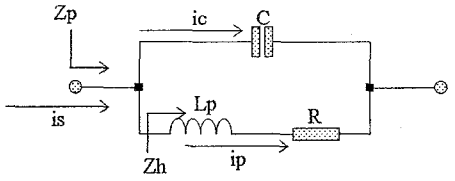

Fig.2: The parallel circuit determines the ratio of the currents i_p/i_s.

The Quality Factor.

The quality factor Q can be defined, in general, for a second order system, as,

$$Q = \frac{\omega_0}{2\alpha} \tag{3}$$

where,

$$\lambda_{1,2} = -\alpha \pm j\sqrt{\omega_0^2 - \alpha^2} \tag{4}$$

are the solutions of the characteristic polynomial of the system. ω_0 is called resonant frequency and α is called damping coeficient.

In agreement with the schematic of the *fig.2*, the parallel circuit impedance is,

$$Z_p(S) = \frac{L_p S + R}{CL_p S^2 + RCS + 1} \tag{5}$$

Due to the presence of *C*, this impedance can only be atacked by current, giving a voltage response *v* led by the following second order equation,

$$\left(CL_p S^2 + RCS + 1\right)v = \left(L_p s + R\right)i \tag{6}$$

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with characteristic polynomial,

$$CL_p\lambda^2 + RC\lambda + 1 = 0 \tag{7}$$

Taking (4) into account, the damping coeficient will be,

$$\alpha_p = \frac{R}{2L_p} \tag{8}$$

and the resonant frequency,

$$\omega_{0p} = \frac{1}{\sqrt{L_p C}} \tag{9}$$

Using (3), the quality factor of the parallel circuit is,

$$Q_p = \frac{1}{R}\sqrt{\frac{L_p}{C}} \tag{10}$$

The Transfer Function H_i.

The transfer function H_i is defined as the ratio between the two currents flowing through the inductors, that is,

$$H_i(\omega) = \frac{i_p(\omega)}{i_s(\omega)} \tag{11}$$

The main virtue of the R.T. circuit arises from the possibility of making a design in which the series inductor current is many times less than the parallel inductor current. So, it is possible to have a great dissipation on the load but low stresses on the switches, in contrast with the series resonant inverter where all the current through the heating coil flows also through the switches. That´s the reason why the study of the module of the transfer function H_i is interesting, as well as the calculation of its maximum.

Starting from (11), it´s easy to reach,

$$H_i(\omega) = \frac{1}{1 - L_p C\omega^2 + jRC\omega} \tag{12}$$

Thus, the module of H_i becomes,

$$\left|H_i(\omega)\right|^2 = \frac{1}{\left(1 - L_p C\omega^2\right)^2 + R^2C^2\omega^2} \tag{13}$$

The calculation of the maximum of (13) is the calculation of the minimum of its denominator, giving,

$$\omega_H_i max = \frac{R}{L_p}\sqrt{Q_p^2 - \frac{1}{2}} \tag{14}$$

Using (14) on (13) we get the maximum value of H_i, i.e., the maximum possible value for the ratio i_p/i_s,

$$H_i max = \frac{2Q_p^2}{\sqrt{4Q_p^2 - 1}} \tag{15}$$

If the following condition is accomplished,

$$Q_p >> 1 \tag{16}$$

then (14) and (15) can be approximated by,

$$\omega_H_i max \approx \frac{1}{\sqrt{L_p C}} \tag{17}$$

$$H_i max \approx Q_p \tag{18}$$

In *fig.3* it´s possible to observe the evolution of the module of H_i as a function of the frequency, as well as the validity of the approximations made in (17) and (18).

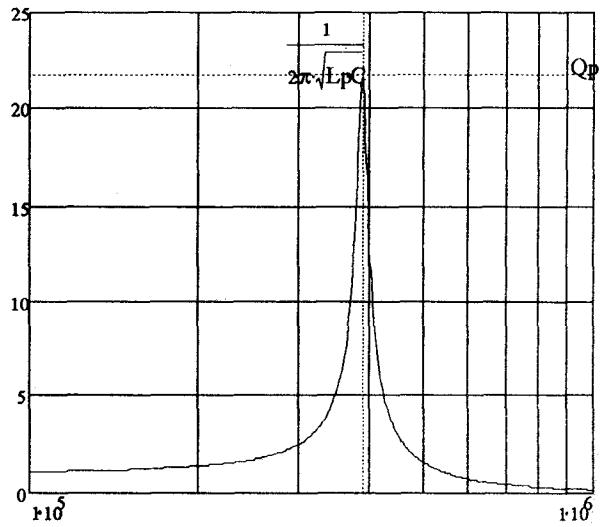

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Fig.3: Curve representation of $|H_i(f)|$ to verify the equations (17) and (18).

On the other hand, it´s easy to prove that the argument of H_i when its module reaches the maximum, it´s approximatly -90 degees, as can be seen in *fig.7*.

THE R.T. CIRCUIT

Power Analysis.

The aim of this seccion is to get a design equation which allows to stablish the maximum power at the desired frequency. For that, it is necesary to get the expression of the frequency at which the maximum power dissipation is obtained.

Admiting the approximation such that all the consumption is due to the first harmonic of the input voltage, the expression of the power, $P(\omega)$, yields,

$$P(\omega)=\frac{v_1^2}{2}\mathrm{Re}\left[\frac{1}{Z(\omega)}\right] \tag{19}$$

so, the power will be maximized when,

$$\mathrm{Re}\left[\frac{1}{Z(\omega)}\right] \tag{20}$$

reaches its maximum value.

Writing $Z(\omega)$ in the form,

$$Z(\omega)=\frac{a+jb}{c+jd} \tag{21}$$

where,

$$a=R\left(1-L_sC\omega^2\right) \tag{22}$$

$$b=\left(L_s+L_p\right)\omega-L_sL_pC\omega^3 \tag{23}$$

$$c=1-CL_p\omega^2 \tag{24}$$

$$d=RC\omega \tag{25}$$

the expression (20) results,

$$\mathrm{Re}\left[\frac{1}{Z(\omega)}\right]=\frac{ca+bd}{a^2+b^2}=\frac{R}{P_6(\omega)} \tag{26}$$

where,

$$P_6(\omega)=K_6\omega^6+K_4\omega^4+K_2\omega^2+R^2 \tag{27}$$

$$K_2=\left(L_s+L_p\right)^2-2R^2L_sC \tag{28}$$

$$K_4=\left(L_sRC\right)^2-2\left(L_s+L_p\right)L_sL_pC \tag{29}$$

$$K_6=\left(L_sL_pC\right)^2 \tag{30}$$

Thus, to find the maximum of (26) consist on finding the minimum of the polynomial P_6, i.e., we have to find the values ω satisfying,

$$\frac{d}{d\omega}P_6(\omega)=0 \tag{31}$$

Making the variable change,

$$x(\omega)=\omega^2 \tag{32}$$

we get,

$$P_6(\omega)=P_3(x) \tag{33}$$

and the equation (31) can be expanded out yielding,

$$\frac{dP_6}{d\omega}=\frac{dP_3}{dx}\cdot\frac{dx}{d\omega}=\frac{dP_3}{dx}\cdot 2\omega=0 \tag{34}$$

Neglecting the solution ω=0, the equation to solve is,

$$\frac{d}{dx}P_3(x)=Dx^2+Ex+F=0 \tag{35}$$

where,

$$D=3\left(L_sL_pC\right)^2 \tag{36}$$

$$E=2\left[\left(L_sRC\right)^2-2\left(L_s+L_p\right)L_sL_pC\right] \tag{37}$$

$$F=\left(L_s+L_p\right)^2-2R^2L_sC \tag{38}$$

giving the following expression of the frequency at which we get the maximum power dissipation,

$$\omega_Pmax=\sqrt{\frac{-E+\sqrt{E^2-4DF}}{2D}} \tag{39}$$

Before use this result as a design equation, it could be useful to make a simplification. Defining,

$$L=\frac{Ls\cdot Lp}{Ls+Lp} \tag{40}$$

$$\omega_0=\frac{1}{\sqrt{LC}} \tag{41}$$

$$Q=\frac{1}{R}\sqrt{\frac{L}{C}} \tag{42}$$

the condition,

$$\boxed{Q>>\sqrt{2}} \tag{43}$$

is enough for the following (44) and (45),

$$Q>>\sqrt{2}>1>\frac{1}{\sqrt{2}}\sqrt{\frac{L}{L_p}} \tag{44}$$

$$Q>>\sqrt{2}>\sqrt{2}\sqrt{\frac{L}{L_p}}\sqrt{\frac{L}{L_s}} \tag{45}$$

Taking into account (44) we get,

$$E\approx-4\left(L_s+L_p\right)L_sL_pC \tag{46}$$

and using (45) we get,

$$F\approx\left(L_s+L_p\right)^2 \tag{47}$$

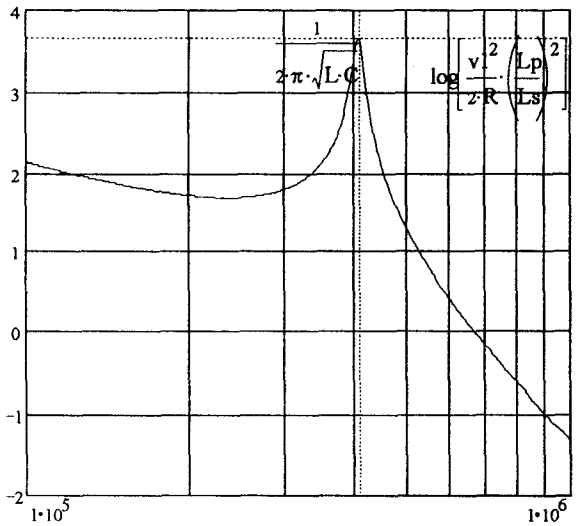
Admiting (46) and (47), ω_Pmax can be approximated by,

$$\boxed{\omega_Pmax \approx \omega_0 = \frac{1}{\sqrt{LC}}} \tag{48}$$

Finally, the maximum power value will be, introducing (48) in (19),

$$\boxed{Pmax \approx \frac{v1^2}{2R}\left(\frac{L_p}{L_s}\right)^2} \tag{49}$$

Fig.4 presents the evolution of the power versus the frequency, showing the validity of the equations (48) and (49).



— log(P)

Fig.4: Aspect of the first harmonic dissipation versus the frequency, and verifying of equations (48) and (49).

The control of the power by frequency changes requires the first one to be a monotonously dependent of the second one. Looking at *fig.4*, the previous condition is accomplished in the range $\omega > \omega_Pmax$. This range will be the working frequency interval of the R.T. circuit under the frequency control law.

On the other hand, substituting (48) on (13), the module of H_i, when the maximum power is transferred to the load, is approximately,

$$H_i_Pmax \approx \frac{Q}{\sqrt{1+\left(Q\frac{L_p}{L_s}\right)^2}} \tag{50}$$

In *fig.5* this last equation is verified.

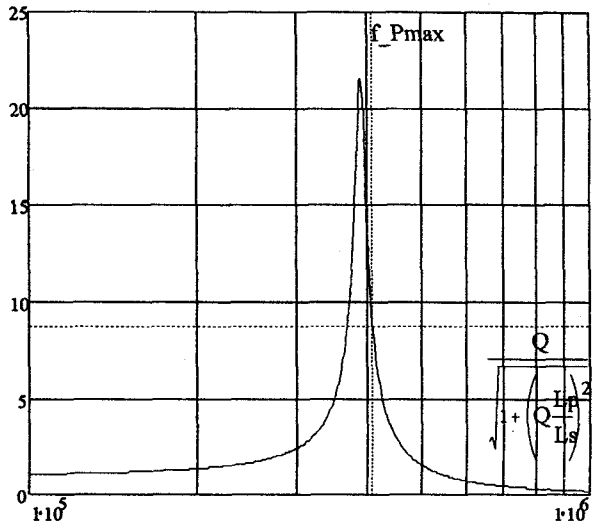


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Fig.5: Value of H_i at the point of maximum power.

Analysis Of The Current i_p.

The maximum value of i_p is obtained when the power reaches its maximum, so,

$$Pmax = \frac{\left(i_p max\right)^2 R}{2} \tag{51}$$

Hence, the maximum current $i_p max$ results,

$$i_p max = \sqrt{\frac{2Pmax}{R}} \tag{52}$$

and using (49) in (52) yields,

$$\boxed{i_p max \approx \frac{v1}{R}\frac{L_p}{L_s}} \tag{53}$$

Analysis Of The Current i_s.

Despite its maximum happens at the frequency which minimizes the module of the R.T.'s impedance, the result of this calculation is not useful as a design equation. Therefore, let's supose $i_s max$ as the value of i_s at the maximum power point, i.e,

$$i_s_Pmax = \frac{i_p max}{H_i_Pmax} \tag{54}$$

Substituting (50) and (53) in the previous equation, we get,

$$\boxed{i_s_Pmax \approx \frac{v1}{R}\left(\frac{L_p}{L_s}\right)^2\sqrt{1+\left(\frac{L_s}{QL_p}\right)^2}} \qquad (55)$$

In *fig.6* we can check out the capability of the R.T. to keep a big ratio between i_p and i_s.

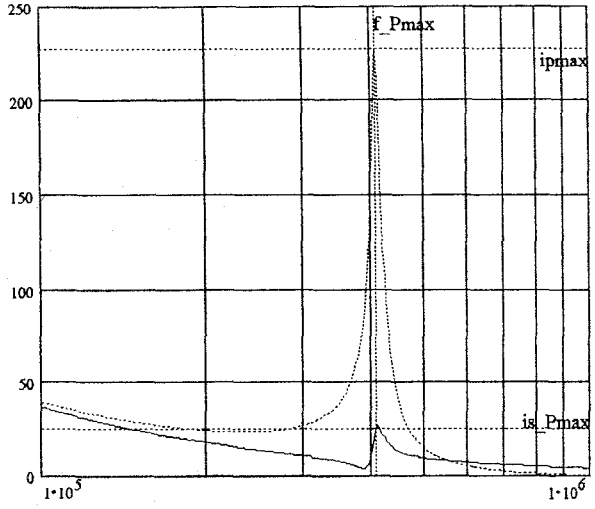

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Fig.6: Amplitudes of the currents i_p and i_s.

The Transfer Function H_V:

We define H_V as the ratio between the capacitor voltage and the first harmonic amplitude of the input voltage,

$$H_v(\omega) = \frac{v_c(\omega)}{v_1} \qquad (56)$$

The importance of the study of this transfer function lies in the fact that its module will allow us to determine the voltage stress in the capacitor, and its phase will be the control variable in the implementation of the control loop.

Starting from,

$$H_v(\omega) = \frac{Z(\omega) - j\omega L_s}{Z(\omega)} \qquad (57)$$

we get,

$$H_v(\omega) = \frac{R + j\omega L_p}{a + jb} \qquad (58)$$

where *a* and *b* are the defined in (22) and (23). Thus,

$$\left|H_v(\omega)\right|^2 = \frac{R^2 + L_p^2\omega^2}{P_6(\omega)} \qquad (59)$$

being P_6 the defined in (27). Using (48) in (59), we get,

$$H_v_Pmax \approx \frac{L_p}{L_s}\sqrt{1+\left(\frac{L_p}{L}\right)^2 Q^2} \qquad (60)$$

and according to the approximation (43),

$$H_v_Pmax \approx \frac{L_p}{L_s}\left(1+\frac{L_p}{L_s}\right)Q \qquad (61)$$

On the other hand, the H_V expression (58) can be also written as,

$$H_v(\omega) = \frac{(R + j\omega L_p)(a - jb)}{P_6(\omega)} \qquad (62)$$

and therefore, the phase of H_V will be the argument of the numerator of (62). Using (48) in the equation above, the phase of H_V yields,

$$\arg\{H_v(\omega_0)\} = \\ = \arg\left\{\frac{-L_s}{L_s + L_p} - jQ\right\} \approx -\frac{\pi}{2} \qquad (63)$$

In *fig.7* the phase of H_V is drawn versus the frequency. It can be seen that the maximum power is dissipated when the phase of H_V equals to -90 degrees, independently of this maximum power value. Moreover, in $\omega > \omega_Pmax$ the power is a monotonously increasing function of the phase of H_V, i.e., the phase of H_V gives an indirect information about the delivered power (normalized at its maximum).

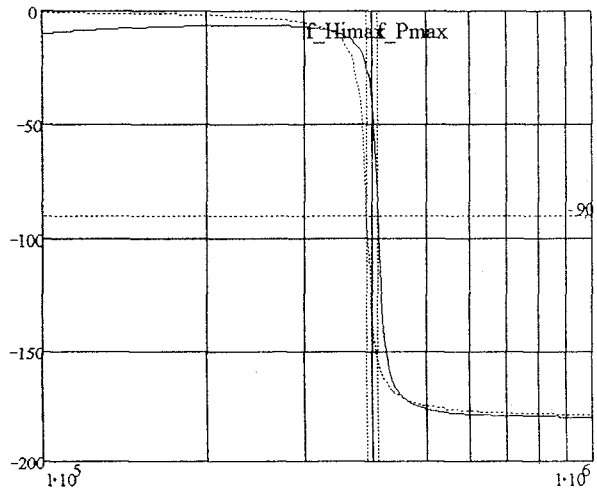

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Fig.7: Phases of H_i and H_V. The phase of H_V is the control variable which must be sensed.

Analysis Of The Voltage v_C:

The maximum voltage in the capacitor will be,

$$v_c_Pmax \approx v_1 \frac{L_p}{L_s}\sqrt{1+\left(\frac{L_p}{L}Q\right)^2} \qquad (64)$$

and using (43) it can be simplified as,

$$\boxed{v_c_Pmax \approx v_1 \frac{L_p}{L_s}\left(1+\frac{L_p}{L_s}\right)Q} \qquad (65)$$

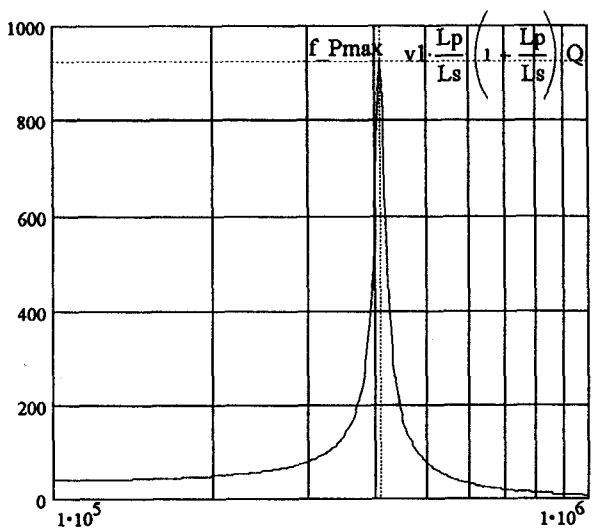

Fig.8: Amplitude of the capacitor voltage.

DESIGN ALGORITHM OF THE R.T. IMPEDANCE

The starting data are:

1) Voltage $\pm Vdc$ applied to the resonant (Vdc).
2) Frequency of maximum power (f_0).
3) Maximum power ($Pmax$).
4) Equivalent estimated resistance of the load (R).
5) Maximum current allowed through the switches ($i_s max$).
6) Maximum voltage allowed in the capacitor ($v_C max$).

Step 1.- Calculate the amplitude of the first harmonic of the input voltage:

$$v_1 = \frac{4 \cdot Vdc}{\pi} \qquad (d\text{-}1)$$

Step 2.- Check the following inequality:

$$i_s max > \frac{2Pmax}{v_1} \qquad (d\text{-}2)$$

If it is not accomplished means that it is not possible to keep the current i_S below its maximum allowed value $i_S max$ satisfying the input power requirements, i.e., it is not possible to carry out the specifications.
The second side of the inequality is the lowest current that it's possible to design with the constrain of the maximum desired power, and it's obtained with an infinite Q, i.e., when i_S and v_1 have the same phase.

Step 3.- Calculate the ratio α between the parallel and series inductance:

$$\alpha = \frac{L_p}{L_s} = \frac{\sqrt{2 \cdot Pmax \cdot R}}{v_1} \qquad (d\text{-}3)$$

Step 4.- Determination of the boundaries of Q:
I_S is monotonously decreasing with Q, so we must take a Q higher than the corresponding to $i_S max$, called $Qmin$,

$$Qmin = \frac{1}{\alpha\sqrt{\left(\frac{i_s max \cdot v_1}{2Pmax}\right)^2 - 1}} \qquad (d\text{-}4)$$

$Qmin$ has always a real solution if the condition of the step 2 is accomplished.
V_C is monotonously increasing function of Q, so we should take a Q lower than the corresponding to $v_C max$, called $Qmax$,

$$Qmax = \frac{v_c max}{v_1 \cdot \alpha \cdot (1+\alpha)} \qquad (d\text{-}5)$$

If the unequality Qmax>Qmin is not accomplished, that means that the $v_C max$ demanded is too restrictive, and the design has only solution if a higher $v_C max$ is allowed.

Step 5.- Selection of Q:
- Should be close to $Qmax$ if we want to decrease the current stress in the switches.
- Should be close to $Qmin$ if we want to decrease the voltage stress in the capacitor.

Most commonly is preferable to choose a Q close to its maximum to reduce the current through the switches.

Step 6.- Calculation of C, L_p, L_S:

$$C = \frac{1}{Q \cdot R \cdot 2\pi \cdot f_0} \qquad (d\text{-}6)$$

$$L_p = \frac{Q \cdot (\alpha+1) \cdot R}{2\pi \cdot f_0} \qquad (d\text{-}7)$$

$$L_s = \frac{L_p}{\alpha} \qquad (d\text{-}8)$$

Design Example.

We take, as an input data:

Vdc=310v	f_0=200Khz
$Pmax$=3Kw	R=0.18Ω
$i_s max$=20A	$v_c max$=1000v

Step 1:

$$v_1 = \frac{4 \cdot V_{dc}}{\pi} = \frac{4 \cdot 310}{\pi} = 394.7$$

Step 2:

$$i_s max = 20 > \frac{2Pmax}{v_1} = \frac{2 \cdot 3000}{394.7} = 15.2$$

That means that the current i_s could be reduced until 15.2A with a big enough quality factor.

Step 3:

$$\alpha = \frac{\sqrt{2 \cdot Pmax \cdot R}}{v_1} = \frac{\sqrt{2 \cdot 3000 \cdot 0.18}}{394.7} = 0.083$$

Step 4:

$$Qmin = \frac{1}{\alpha\sqrt{\left(\frac{i_s max \cdot v_1}{2Pmax}\right)^2 - 1}} =$$

$$= \frac{1}{0.083\sqrt{\left(\frac{20 \cdot 394.7}{2 \cdot 3000}\right)^2 - 1}} = 14.09$$

$$Qmax = \frac{v_c max}{v_1 \cdot \alpha \cdot (1+\alpha)} =$$

$$= \frac{1000}{394.7 \cdot 0.083 \cdot 1.083} = 28.1$$

Step 5:

We choose Q=28 to minimize the current i_s.

Step 6:

Finally, the calculation of the three reactive elements results in,

$$C = \frac{1}{Q \cdot R \cdot 2\pi \cdot f_0} = \frac{1}{28 \cdot 0.18 \cdot 2 \cdot \pi \cdot 200000} = 157.9nF$$

$$L_p = \frac{Q \cdot (\alpha + 1) \cdot R}{2\pi \cdot f_0} = \frac{28 \cdot 1.083 \cdot 0.18}{2 \cdot \pi \cdot 200000} = 4.3\mu F$$

$$L_s = \frac{L_p}{\alpha} = \frac{4.3}{0.083} = 51.8\mu F$$

IV. CONCLUSION

A complete steady-state frequency analysis of the LLC resonant tank for induction heating has been performed in this paper, providing the analitical expressions and approximations needed to solve the design problem.

This study has been followed lately by the small-signal analysis, and the control design and stability evaluation, completing the theorical study of this inverter.

The circuit is going to be built soon, with the expectation of a promising results which make it more advantageous than the SRI (Series Resonant Inverter) for induction heating applications.

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